

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 03

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• $(120 + 95) - (2 + 45) = 168 \text{ t ha}^{-1}$	1
01.2	• Annual ploughing aerates the soil so aerobic decomposers break down dead organic matter faster, releasing CO₂ • Harvest removes biomass so less dead organic matter is returned; pasture roots and litter constantly add organic matter	2
01.3	<i>Any 2 from:</i> • Low / zero tillage • Adding manure, compost or mulch • Growing cover crops / leaving crop residues • Converting to long-term grass or tree crops in rotation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-151; AQA spec 3.2.4.2.	
02.1	• B	1
02.2	• Number all possible sampling sites (eg on a grid map) • Select the sites using a random number generator / table; (regular shape: grid coordinates from random numbers)	2
02.3	• Sites may be chosen for convenience or because they appear to support (or dismiss) the hypothesis • So the sample does not represent the whole area and conclusions are invalid	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	

Question	Marking guidance	Marks
03.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Fossil fuel technologies: secondary / tertiary recovery raising recovery from 20% to 60%; directional drilling and subsea wells; ROVs; fracking for tight oil and gas; tar sands / oil shales; coal gasification / liquefaction; enhanced gas recovery; methane hydrates; CCS (pre / post combustion, storage) • Limits: still finite (discoveries declining since the 1980s); rising energy and financial cost of unconventional sources; environmental impacts (aquifers, water use, tremors, land take, habitat damage); CO₂ and other pollution; CCS unproven at scale and impractical for vehicles • Nuclear: new uranium extraction (polymer adsorption from seawater, phosphate mining, coal ash); molten salt reactors; plutonium breeder and thorium reactors increasing fuel use many-fold; fusion as a long-term prospect • Limits: cost overruns, decommissioning, waste, public opposition, proliferation, embodied energy of low-grade uranium; fusion not commercial • Sustainability criteria: maintaining supply without unacceptable environmental, economic or social impact; comparison with renewables and energy conservation; energy gap argument • Judgement: technologies can extend supplies and reduce some impacts but cannot make non-renewables sustainable indefinitely; role as transition / base-load alongside renewables	9
03.2	• 80-90%	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 183-196; AQA spec 3.3.3-3.3.4.1.	
04.1	• The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically)	1

Question	Marking guidance	Marks
04.2	<i>Any 4 from:</i> • Light: UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions • Temperature: higher temperatures increase the rate of chemical breakdown / microbial decomposition • Soil microbes / oxygen: aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils • pH / presence of other chemicals may increase or decrease the rate of degradation • Adsorption onto clay / organic matter may protect from degradation and increase persistence	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
05.1	• Production prevention: desulfurisation of fuels before combustion; low-temperature (fluidised bed) combustion to prevent NO_x; smokeless fuels • Prevention of release: electrostatic precipitators / bag filters / cyclones for smoke; catalytic converters; flue gas desulfurisation; bund walls; landfill liners • Post-release remediation: oil spill clean-up (booms, skimmers, dispersants, bioremediation); phytoremediation of metal-contaminated land; liming of acidified lakes • Alternative processes: electric vehicles instead of petrol / diesel; renewable energy instead of fossil fuels; pyrethroids instead of organochlorines; fuel cells instead of internal combustion engines	4
05.2	• Remediation is often incomplete / cannot recover all the pollutant (eg oil spills, dispersed pollutants), so some damage still occurs • Prevention avoids the damage entirely and is usually cheaper than clean-up / avoids polluter-pays costs; moving from post-production treatment to non-release is a circular economy principle	2
05.3	• Satellites detect and track the extent / movement of oil slicks over large areas, quickly and repeatedly, so clean-up can be directed and polluters identified • GPS tracking monitors ship positions so navigation errors / collisions / groundings are reduced and offshore shipping routes are enforced • Polymers with a high surface area adsorb metal ions from contaminated water / effluent so they can be removed and recovered • Plants that absorb heavy metals (hyperaccumulators) are grown on contaminated land, harvested and disposed of / metals recovered, cleaning the soil	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 244, 303-304; AQA spec 3.4.3.1, 3.4.3.3.	
06.1	• UV light splits diatomic oxygen (O₂) into two monatomic oxygen atoms (O) • Monatomic oxygen reacts with diatomic oxygen to form ozone (O₃) • UV light also splits ozone into O₂ and O; the reactions form a dynamic equilibrium which absorbs UV (converting it to chemical energy / heat)	3

Question	Marking guidance	Marks
06.2	• Useful: boiling points near ambient temperature so easily liquefied under low pressure (aerosol propellants, fridge / air-conditioning coolants); dissolve grease but do not damage electrical components (solvents); non-flammable; non-toxic • Persistence: chemically stable / unreactive in the troposphere so they remain long enough to be carried up to the stratosphere • Dissociation by UV: in the stratosphere higher UV levels break the carbon-chlorine bonds releasing chlorine free radicals • Chlorine reacts with monatomic oxygen (and ozone), preventing ozone formation; chlorine is regenerated so one atom destroys many ozone molecules; bromine / iodine act similarly	6
06.3	• UVB energy is absorbed by cells and breaks up biological molecules / DNA causing mutations, skin cancer, cataracts, sunburn (human health) • Damage to leaf tissue and reduced photosynthesis / crop yields (plants) • Damage to marine algae, phytoplankton, corals and planktonic larvae (marine organisms), reducing productivity and food webs	3
06.4	• An area where the total ozone column falls below 220 Dobson Units (normal is about 300 DU) - a level never recorded before CFC pollution	1
06.5	• Concentration varies with altitude and the stratosphere is not normally sampled; Dobson Units measure the total column thickness instead	1
06.6	• The British Antarctic Survey at Halley detected higher UV at ground level, showing ozone above had decreased (satellites above the stratosphere cannot measure downward UV directly)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-109; AQA spec 3.2.1.3.	
07.1	• Negative feedback: a change causes other changes that reduce the rate or size of the original change, re-establishing equilibrium • Positive feedback: a change causes other changes that increase the original change	2
07.2	• Increased low-level cloud: higher temperatures increase evaporation and condensation, producing more cloud; clouds have a high albedo so more sunlight is reflected and warming is reduced • Increased photosynthesis: higher temperatures and CO₂ increase photosynthesis, removing CO₂; if stored in woody tissue atmospheric CO₂ rises less	4
07.3	• Permafrost: frozen waterlogged Arctic soils contain dead organic matter that decomposed anaerobically producing methane trapped in the frozen ground • Warming thaws the permafrost, releasing methane (a powerful greenhouse gas) and allowing further decomposition, causing more warming • Methane hydrate: methane from decomposed organic matter in deep sea sediments is held as a solid under high pressure and low temperature • If deep water warms the hydrate melts, releasing methane into the atmosphere and causing further warming	4

Question	Marking guidance	Marks
07.4	<i>Any 3 from:</i> • Reduced albedo as snow / ice melts • Ocean acidification reducing coral survival and carbon sequestration • Increased forest and peat fires releasing CO₂ • Faster decomposition of soil organic matter releasing CO₂ • Increased water vapour (a greenhouse gas) in warmer air • Increased cirrus (high) cloud which traps infrared	3
07.5	• A point at which natural processes triggered by human-caused warming become self-sustaining / unstoppable • So that stopping the original human activity would no longer stop climate change (eg runaway permafrost melt, methane hydrate release, forest fires)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-104, 371-372; AQA spec 3.2.1.2, 3.6.1.	
08.1	• Oxygen	1
08.2	• A change to one process causes other processes to alter as a direct result (chains of linked processes / feedback) • There is still a lot that is not understood about how atmospheric processes work	2
08.3	• The surface warmed in the tropics heats the air above; warm winds (south-westerlies from the Caribbean) carry heat energy to higher latitudes • Winds carry water vapour (evaporated from warm oceans) to areas that would otherwise receive little precipitation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-88; AQA spec 3.2.1.1.	
09.1	• Screens (metal grills) trap floating / suspended non-biodegradable items (paper, plastic) which are landfilled or incinerated • Grit traps: the channel widens so flow slows and stones / road grit sink out because there is not enough kinetic energy to carry them	2
09.2	• The effluent stands in large tanks so the faecal / organic solids sink and are removed as sludge • This removes over 95% of the organic matter from the fluid so the risk of deoxygenation of the river is greatly reduced	2
09.3	• Aerobic bacteria digest / break down the remaining dissolved and suspended organic matter • Air (or pure oxygen) is mixed in by paddle wheels or air stones so oxygen shortage does not limit the aerobic process • The suspended bacteria collected in the secondary sedimentation tank are the 'activated sludge'; returning them maintains a large bacterial population so digestion is rapid	3
09.4	• Effluent from secondary treatment may still contain dissolved inorganic nutrients (phosphates, nitrates) that cause eutrophication, and some bacteria • Needed where the discharge site is ecologically sensitive or used by many people (bathing water, drinking water abstraction) • Phosphate stripping with iron(III) sulfate; micro-strainers (very fine sieves) then UV light or chlorine to kill remaining bacteria	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 269-272; AQA spec 3.4.3.2.9.	

Question	Marking guidance	Marks
10.1	• $4.2 / 0.7 = 6$	1
10.2	• Long summer days (almost 24 hours of daylight) so the total energy received is high, even though the Sun is low • In winter days are very short / the Sun does not rise, so light levels are extremely low	2
10.3	• Reflectors can only focus parallel rays of direct sunlight onto the absorber • Clouds, smoke and dust scatter the light so it cannot be focused and intensified	2
10.4	• Very large areas of collectors are needed to harness significant amounts of energy (and the changing angle of incidence reduces it further)	1
10.5	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Problems: intermittency (daily and seasonal cycles), unpredictable cloud cover, low energy density, changing angle of incidence, land requirements, water for cleaning in deserts, toxic wastes in manufacture (silicon tetrachloride, cadmium), efficiency loss at high temperature • Photothermal: evacuated tubes reduce conduction / convection losses; thermal stores (water, sand, concrete, molten salt) allow later use; passive solar architecture, brise soleil, light tubes; heat pumps deliver up to four times the energy used • Photovoltaic developments: multijunction cells absorbing more wavelengths, anti-reflective / moth-eye textured surfaces (smooth cells reflect 30%), PV-thermal hybrids cooling the cell, transparent PV in windows, heliostats keeping the optimum angle, self-cleaning nanohydrophobic surfaces • Concentrating solar power with molten-salt storage to 550 °C so steam turbines run when needed - overcomes intermittency • Integration: grid-connected solar farms, rooftop space avoiding land conflicts, batteries, pumped storage, links to energy storage and demand management • Evaluation: strong in sunny deserts / low latitudes, weaker in high latitudes in winter; cost falling; manufacturing impacts; no single technology solves all problems; justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 199-203; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
11.1	Explain how feedback mechanisms and tipping points influence the rate of global climate change and discuss the implications for the strategies needed to control it. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Negative feedback: increased low-level cloud (albedo), increased photosynthesis and sequestration; hydrological cycle evaporation-precipitation • Positive feedback: melting permafrost (methane), reduced albedo of snow / ice / sea ice, methane hydrate release, ocean acidification and coral loss, forest and peat fires, soil decomposition, water vapour, cirrus clouds • Tipping points: self-sustaining processes; why control must precede them; examples • Implications: urgency and precautionary principle; need for deep cuts in greenhouse gases (energy, transport, buildings, agriculture, forestry, CCS) rather than only adaptation; geoengineering risks • Difficulty of prediction because feedbacks interact at different scales; models; international agreements and per capita footprints • Judgement on whether current strategies are adequate Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-107, 371-372; AQA spec 3.2.1.2, 3.6.1.	25

Question	Marking guidance	Marks
11.2	Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Scale of the problem: how energy is used (transport, buildings, industry, water); links between affluence and per capita use • Transport: bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles • Buildings: orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour • Industry: heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management • Limits: rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour • Evaluation with quantitative examples; comparison with the cost of new supply; conclusion that conservation is the cheapest first step but cannot alone meet demand growth Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.	25